PC accessibility crisis

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Abstract

Since late 2025, the global technology sector has faced an acute memory crisis characterized by explosive price inflation in Dynamic Random Access Memory (DRAM). Driven by enterprise demand for High-Bandwidth Memory (HBM) and server DRAM in Artificial Intelligence (AI) data centers, major manufacturers have reallocated manufacturing capacity away from consumer hardware. This paper synthesizes market statistics from Gartner, GSMA, and ITU to analyze the structural mechanics of component cost inflation, its impact on entry-level PC and smartphone markets, and the resulting widening of the global digital divide.

1 Introduction and Component Inflation

The semiconductor industry experienced a fundamental shift beginning in October 2025. Unprecedented capital expenditure in AI infrastructure created an insatiable demand for HBM and enterprise-grade server DRAM. A single AI server requires between 8 and 10 times the memory capacity of a standard enterprise server, incentivizing memory fabricators to prioritize high-margin enterprise orders over consumer-grade DDR5 components.

Contract pricing for standard DDR5 16Gb chips surged from \$8.70 in October 2025 to \$48.00 in September 2026—representing a price surge of over 450% (with spot market spikes reaching up to 600%). According to market research from Gartner, PC memory expenses are projected to reach a peak of 23% of the total bill-of-materials (BOM) in 2026, up from 16% in 2025. Combined DRAM and SSD cost increases are expected to reach 130% by the end of 2026.

2 Market Contraction and Segment Erasure

The rapid rise in component costs has severely eroded margins for budget hardware manufacturers. Vendors are increasingly unable to absorb component inflation, leading directly to the contraction of consumer shipments and the extinction of low-cost hardware tiers.

Key global market impact and forecast metrics (2026–2028):

- **DDR5 16Gb contract price increase:** +451% (\$8.70 to \$48.00)
- **Peak memory share of PC BOM (2026):** 23% (up from 16% in 2025)
- **Global PC shipment change (2026 vs 2025):** −10.4%
- **Global smartphone shipment change (2026 vs 2025):** −8.4%
- **Average PC retail price increase (2026):** +17%
- **Average smartphone retail price increase (2026):** +13%
- **Extinction horizon for sub-\$500 entry PC market:** Year 2028 (Gartner forecast)

Gartner explicitly predicts that the **sub-$500 entry-level PC market will disappear by 2028**, as manufacturers pivot exclusively toward premium and enterprise offerings. Consequently, worldwide PC shipments are expected to shrink by 10.4% and smartphone shipments by 8.4% in 2026. To adapt to higher replacement costs, business users are extending PC lifespans by an average of 15%, while consumers are delaying upgrades by 20%.

3 Socio-Economic Consequences and Digital Inclusion

Hardware inflation poses a critical challenge to global digital inclusion initiatives, particularly in developing economies where entry-level hardware serves as the primary gateway to online services, education, and financial systems.

- **Sub-Saharan Africa Handset Burden:** GSMA reporting indicates that an entry-level smartphone currently absorbs **76% of the average monthly income** for the bottom 40% of the population in Sub-Saharan Africa. Sub-$100 smartphone shipments in the region dropped by 34% in Q2 2026.
- **Global Connectivity Divide:** ITU *Facts and Figures 2026* data shows that **2 billion people remain completely offline**. While internet usage reaches 93% in high-income nations, it remains at just 32% in low-income countries.
- **Financial Barriers:** In Least Developed Countries (LDCs), basic mobile broadband connectivity represents a median of 6.1% of Gross National Income (GNI) per capita, far exceeding the UN Broadband Commission's 2% target.

4 Conclusion

The structural reallocation of DRAM wafer capacity toward AI data centers has created widespread hardware inflation across consumer markets. Without policy intervention or targeted subsidies for basic educational hardware, rising component costs will continue to push entry-level computing out of reach for millions, threatening to exacerbate global economic inequality and broaden the digital divide.

5 Source

- https://www.gartner.com/en/newsroom/press-releases/2026-02-26-gartner-says-surging-memory-costs-will-reduce-global-pc-and-smartphone-shipments-in-2026
- https://www.itu.int/en/mediacentre/Pages/PR-2026-11-03-Facts-and-Figures.aspx
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